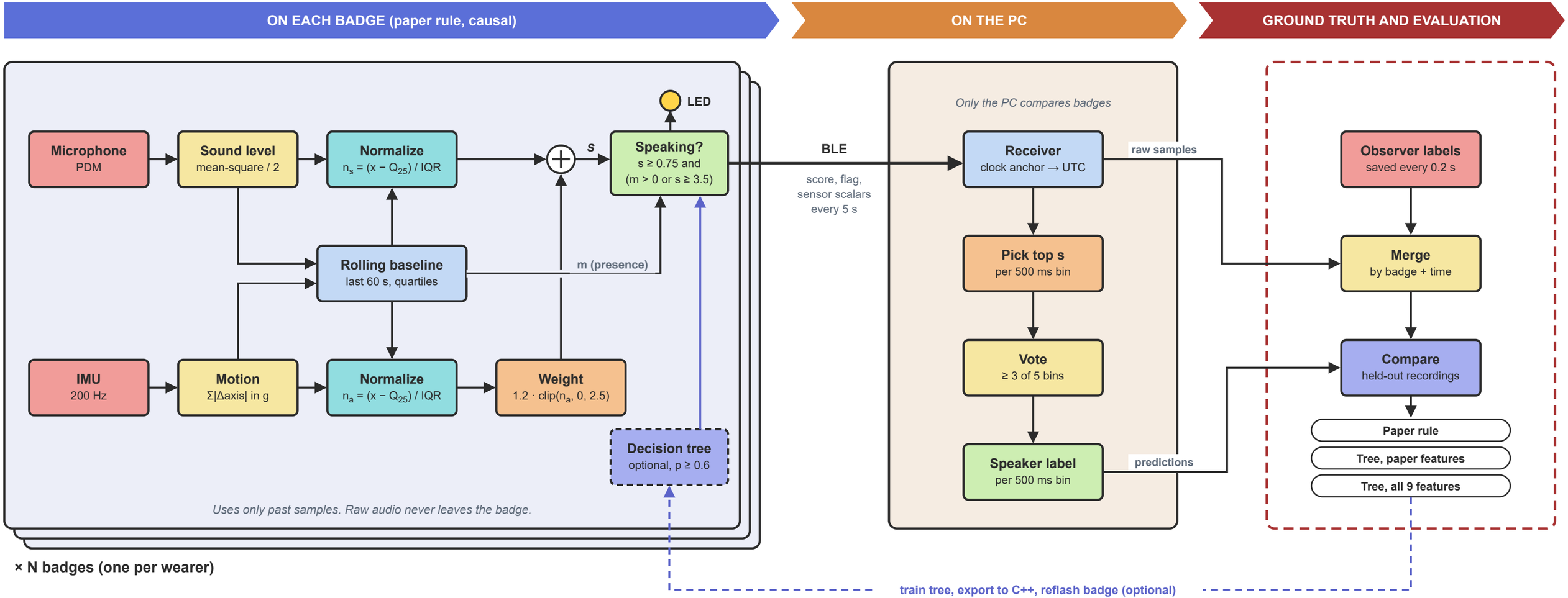

ON EACH BADGE (paper rule, causal)
ON THE PC
GROUND TRUTH AND EVALUATION
Microphone
PDM
Sound level
mean-square / 2
Normalize
$n_s = (x - Q_{25}) / IQR$
s
Speaking?
s ≥ 0.75 and
(m > 0 or s ≥ 3.5)
LED
Rolling baseline
last 60 s, quartiles
m (presence)
IMU
200 Hz
Motion
Σ|Δaxis| in g
Normalize
$n_a = (x - Q_{25}) / IQR$
Weight
$1.2 \cdot clip(n_a, 0, 2.5)$
Decision tree
optional, p ≥ 0.6
Uses only past samples. Raw audio never leaves the badge.
× N badges (one per wearer)
BLE
score, flag,
sensor scalars
every 5 s
Only the PC compares badges
Receiver
clock anchor → UTC
Pick top s
per 500 ms bin
Vote
≥ 3 of 5 bins
Speaker label
per 500 ms bin
raw samples
predictions
Observer labels
saved every 0.2 s
Merge
by badge + time
Compare
held-out recordings
Paper rule
Tree, paper features
Tree, all 9 features
train tree, export to C++, reflash badge (optional)